

Tableau Fields: Normal, Hierarchical & Auto-Generated Fields

1. Normal Fields

Definition:

Normal fields are the fields that already exist in your source data. Tableau imports these fields and allows you to use them for analysis.

Example

Suppose your Excel data looks like this:

ORDER ID	CUSTOMER NAME	REGION	SALES	QUANTITY
101	Amit	West	5000	5
102	Priya	East	3000	3
103	Rahul	West	7000	7

Here, **Order ID**, **Customer Name**, **Region**, **Sales**, and **Quantity** are normal fields.

You can drag:

- **Region** → Columns
- **Sales** → Rows
- **Customer Name** → Detail

Tableau uses the original values from the data source.

Key point

Normal fields = Fields that come directly from your dataset.

2. Hierarchical Fields

Definition:

Hierarchical fields allow you to organize related dimensions into levels so that you can **drill down or drill up** in a visualization.

A common example is a geographical hierarchy:

Country → **State** → **City**

Another example:

Category → **Sub-Category** → **Product Name**

Example

Suppose you have:

COUNTRY	STATE	CITY	SALES
INDIA	Maharashtra	Pune	50,000
INDIA	Maharashtra	Mumbai	70,000
INDIA	Gujarat	Ahmedabad	40,000

You can create a hierarchy:

Geography

- Country
 - State
 - City

When you put the hierarchy into a Tableau view, you can click + to drill down:

India

↓

Maharashtra, Gujarat

↓

Pune, Mumbai, Ahmedabad

You can click – to move back up.

Another example

For a sales dataset:

Product Hierarchy

- Category
 - Sub-Category
 - Product Name

For example:

Technology → Phones → iPhone 15

Key point

Hierarchy = A parent-child structure that lets you drill down from a higher level to a more detailed level.

3. Auto-Generated Fields

Definition:

Auto-generated fields are fields that **Tableau creates automatically** based on the data source or the analysis you perform.

These fields are not necessarily present as separate columns in your original dataset.

Common examples

A. Measure Names

Tableau automatically creates **Measure Names**.

For example, your dataset may contain:

- Sales
- Profit
- Quantity

Tableau can automatically create **Measure Names** containing:

- Sales
- Profit
- Quantity

You can use Measure Names to display multiple measures in a visualization.

B. Measure Values

Measure Values is another automatically generated field.

It contains the actual values of multiple measures.

For example:

MEASURE NAME	MEASURE VALUE
SALES	500,000
PROFIT	100,000
QUANTITY	2,500

You can use **Measure Names + Measure Values** to show several measures in the same chart.

C. Generated Date Fields

When Tableau recognizes a date field, it can provide different date levels automatically.

For example:

Order Date

Tableau can allow you to work with:

- Year
- Quarter
- Month
- Week
- Day

So from one field, **Order Date**, you can analyze sales by year, month, or day.

D. Latitude and Longitude

When Tableau recognizes geographic data, it can automatically generate:

- **Latitude**
- **Longitude**

These are used to create maps.

For example:

City = Pune

Tableau can use geographic information to determine the appropriate latitude and longitude for mapping.

Key point

Auto-generated fields = Fields Tableau creates or provides automatically to help with analysis and visualization.

Quick Comparison

TYPE	MEANING	EXAMPLE
NORMAL FIELD	Comes directly from the source data	Sales, Region, Customer Name
HIERARCHICAL FIELD	Organizes fields into drill-down levels	Country → State → City
AUTO-GENERATED FIELD	Created/provided automatically by Tableau	Measure Names, Measure Values, Latitude, Longitude

Simple way to remember

Think of a **Sales dataset**:

Normal:

Sales, Customer, Region

Hierarchy:

Country → State → City

Auto-generated:

Measure Names, Measure Values, Latitude, Longitude